Manas Deshpande

manasdeshpande51@gmail.com | +91-9321862044 | GitHub | LinkedIn | Portfolio

Summary

Electronics & Computer Science undergraduate specializing in Machine Learning, Computer Vision, NLP, and Retrieval-Augmented Generation (RAG). Experienced building and deploying full-stack AI systems using Python, PyTorch/TensorFlow, FastAPI, and vector databases, including production RAG infrastructure and real-time computer vision pipelines.

Education

Vidyalankar Institute of Technology 2023 – 2027
Bachelor of Engineering – Electronics & Computer Science; CGPA: 7.69/10 *Mumbai, Maharashtra*

Technical Skills

- **Languages:** Python, SQL, JavaScript, HTML, CSS
- **ML/DL:** PyTorch, TensorFlow, Scikit-learn, OpenCV, MediaPipe, CNN, U-Net, YOLOv8
- **Generative AI / NLP:** RAG, LangChain, ChromaDB, FAISS, SentenceTransformers, LLMs (Claude, GPT-4, Gemini), Prompt Engineering
- **Backend & Databases:** FastAPI, Flask, REST APIs, React, Next.js, Streamlit, PostgreSQL, MySQL
- **DevOps & Cloud:** Docker, Git, GitHub Actions, GCP, Azure, Hugging Face Spaces

Experience

Trinetrini Quantum Pvt. Ltd. – AI/ML Intern Mar 2026 – Apr 2026
PELS: Prompt Evaluation & Learning System (RAG Platform) *Navi Mumbai, Maharashtra*

- Built the initial RAG prototype (Jupyter) for PELS, a six-criterion (C1–C6) rubric-based prompt evaluation engine, establishing the proof-of-concept for semantic retrieval and LLM-based scoring.
- Assisted in backend development and scoring-pipeline refinement; integrated multi-team datasets into a unified ChromaDB vector store of 265+ annotated prompts using SentenceTransformer (all-MiniLM-L6-v2) embeddings.
- Authored technical documentation and delivered stakeholder presentations for the production RAG platform (FastAPI + Anthropic Claude Haiku), Dockerized and deployed on Hugging Face Spaces, supporting a live MVP integration on an ed-tech certification platform.

Projects

SignBridge AI GitHub
React, FastAPI, MediaPipe, TensorFlow, WebSockets, Supabase PostgreSQL

- Built a real-time sign-language platform: camera feed → MediaPipe landmark extraction → TensorFlow gesture classification (trained on the Kaggle ASL Alphabet dataset – 87K images, 29 classes) → word/sentence builder, streamed over WebSockets at 30 FPS.
- Added prediction stabilization and confidence tracking to reduce flicker/false positives, backed by a FastAPI + SQLAlchemy service with JWT auth and a Supabase PostgreSQL store for persistent conversation history.

Object Detection & Surveillance System GitHub
YOLOv8, OpenCV, FastAPI, Next.js

- Built a real-time detection pipeline (capture → YOLOv8 detection → ID tracking → counting → movement analysis) at 30 FPS on a self-trained YOLOv8/COCO (80-class) model.
- Designed a modular FastAPI backend exposing `GET /api/detections` (per-class counts, tracked objects, movement data) and `GET /api/stream` for live MJPEG video.
- Built a Next.js (App Router) live dashboard rendering the MJPEG stream alongside real-time detection and entry/exit counts.

Open-Source Contributor – voss-labs/verp Pull Request #42
Next.js, TypeScript, Zod, ExcelJS, REST API

- Shipped a merged feature (PR #42, closing Issue #13) enabling bulk CSV/Excel student onboarding for a college ERP, eliminating impractical one-by-one entry for admins managing hundreds of student records.
- Designed a three-step import workflow (Upload → Preview → Result) with fully client-side parsing via ExcelJS, so no data reaches the server until explicitly confirmed.
- Engineered the `POST /api/students/import` endpoint with Zod schema validation, dual duplicate-detection (intra-batch and against the database), batch insertion with graceful partial-failure handling, per-row error reporting, and audit logging.

Research Experience

Low-Power Clock-Enable Based UART Architecture *Verilog HDL, Xilinx Vivado*

- Designed a Verilog HDL UART architecture with RTL simulation, FPGA synthesis, and ASIC power optimization in Xilinx Vivado.